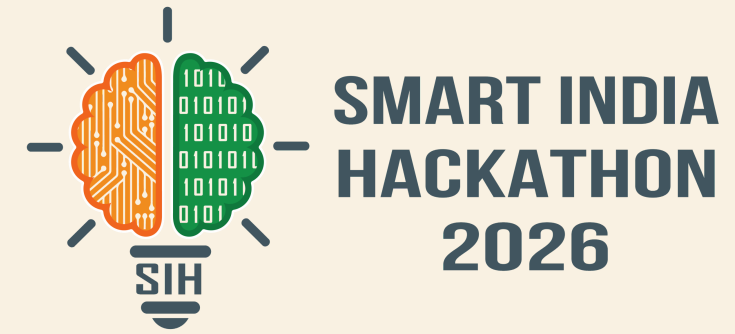


SMART INDIA HACKATHON 2026

ADAPTIVE SPACED RETENTION LEARNING SYSTEM
USING PERSONALIZED FORGETTING MODELS



Problem Statement ID – 26205

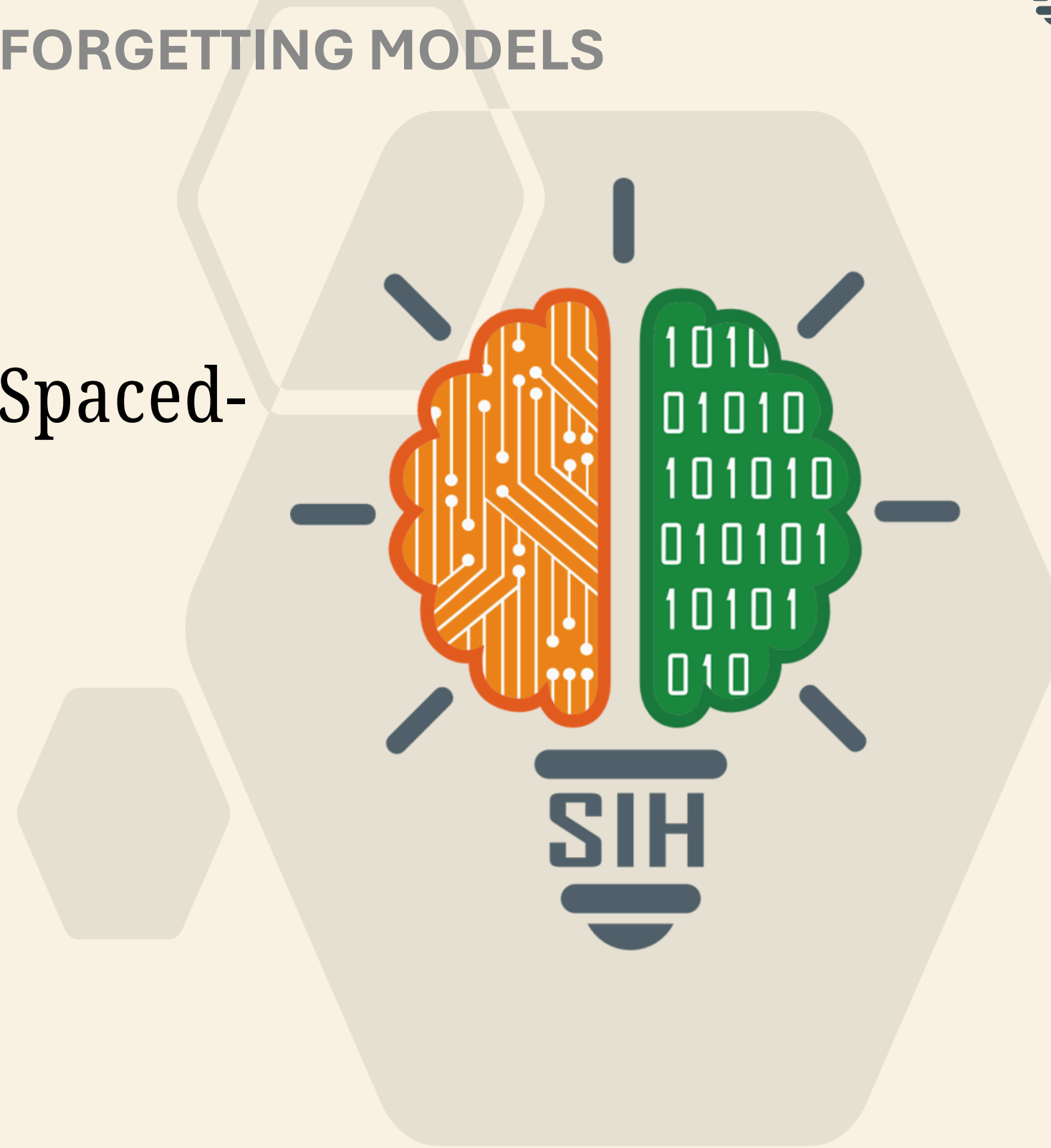
Problem Statement Title – Adaptive Spaced-
Retention Learning System using
Personalized Forgetting Models

Theme – Smart Education

PS Category – Software

Team ID –

Team Name – All Six Not Found



PROPOSED SOLUTION

Detailed Explanation of the Proposed Solution

- MemoRoute allows learners to explain a concept using **voice or text**.
- **LECTOR LLM** evaluates the explanation based on **Correctness, Clarity, and Completeness**.
- The LECTOR score is converted into updates for the personalized forgetting model.
- The model calculates the learner's retention and determines the **next review date**.
- **Exam Mode** prioritizes weak topics based on their retention and the time remaining before the exam.

How It Addresses the Problem

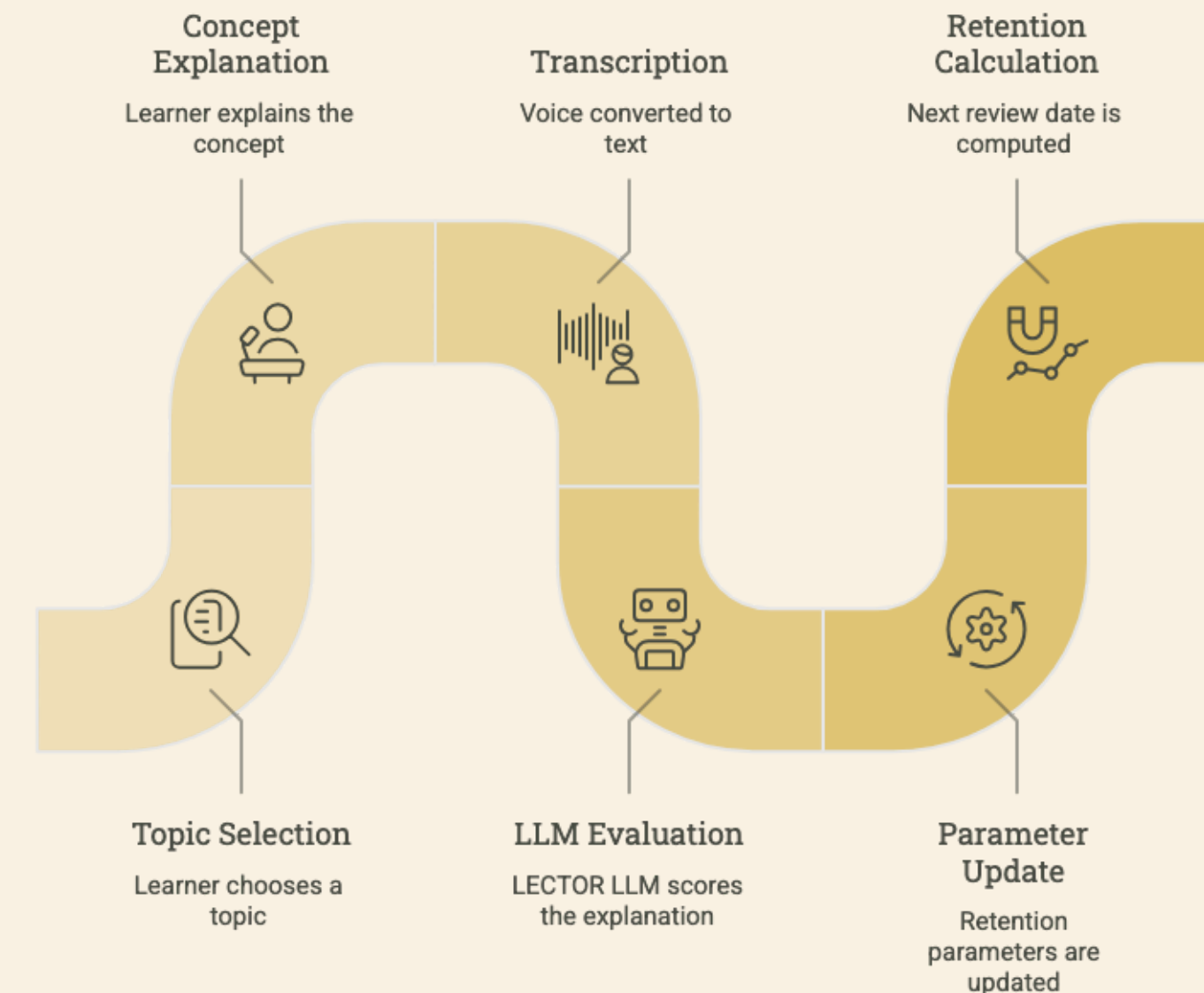
- Checks **actual understanding**, not just right/wrong recall.
- Uses open-ended explanations to identify gaps in understanding.
- Personalizes review intervals according to each learner's performance.
- Works from the **first session** without requiring large historical datasets.
- Helps learners focus on weaker topics before an exam.

PROPOSED SOLUTION

Innovation and Uniqueness

- LECTOR LLM-based comprehension scoring for open-ended explanations.
- Combines the Feynman Technique with Spaced Repetition.
- Uses comprehension scores to drive a personalized forgetting model.
- Provides explainable and adaptive review scheduling.
- Exam Mode dynamically reprioritizes topics based on retention strength.

MindEcho Rescheduling Pipeline

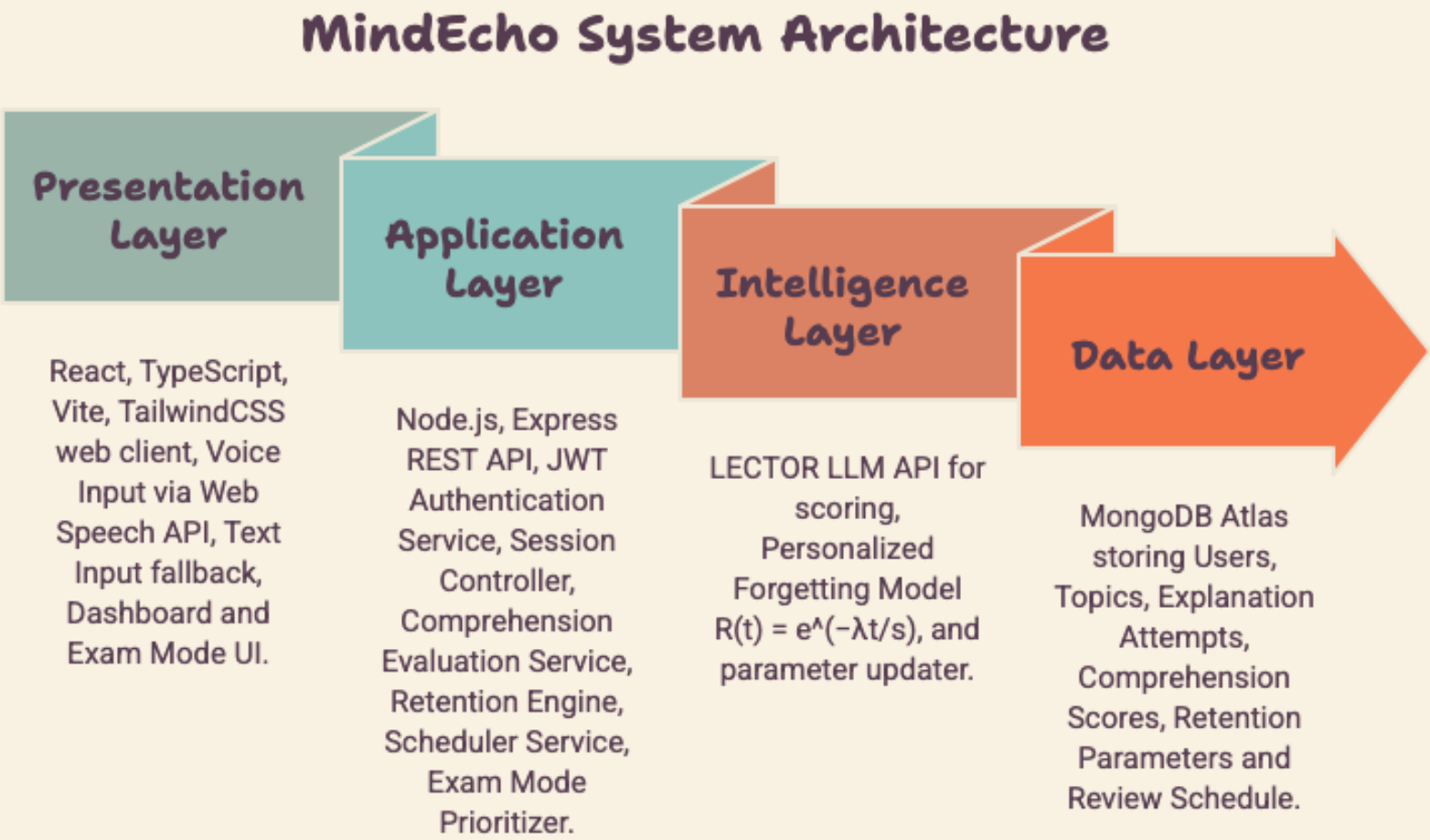


End-to-End Request Flow (best for "how it works")



TECHNOLOGIES

- **Frontend:** React + TypeScript + Vite + TailwindCSS
- **Backend:** Node.js + Express.js
- **Database:** MongoDB / MongoDB Atlas
- **AI:** LECTOR LLM for comprehension scoring and evaluation
- **Voice Input:** Web Speech API
- **Authentication:** JWT
- **Version Control:** Git + GitHub



High-Level System Architecture

FEASIBILITY AND VIABILITY

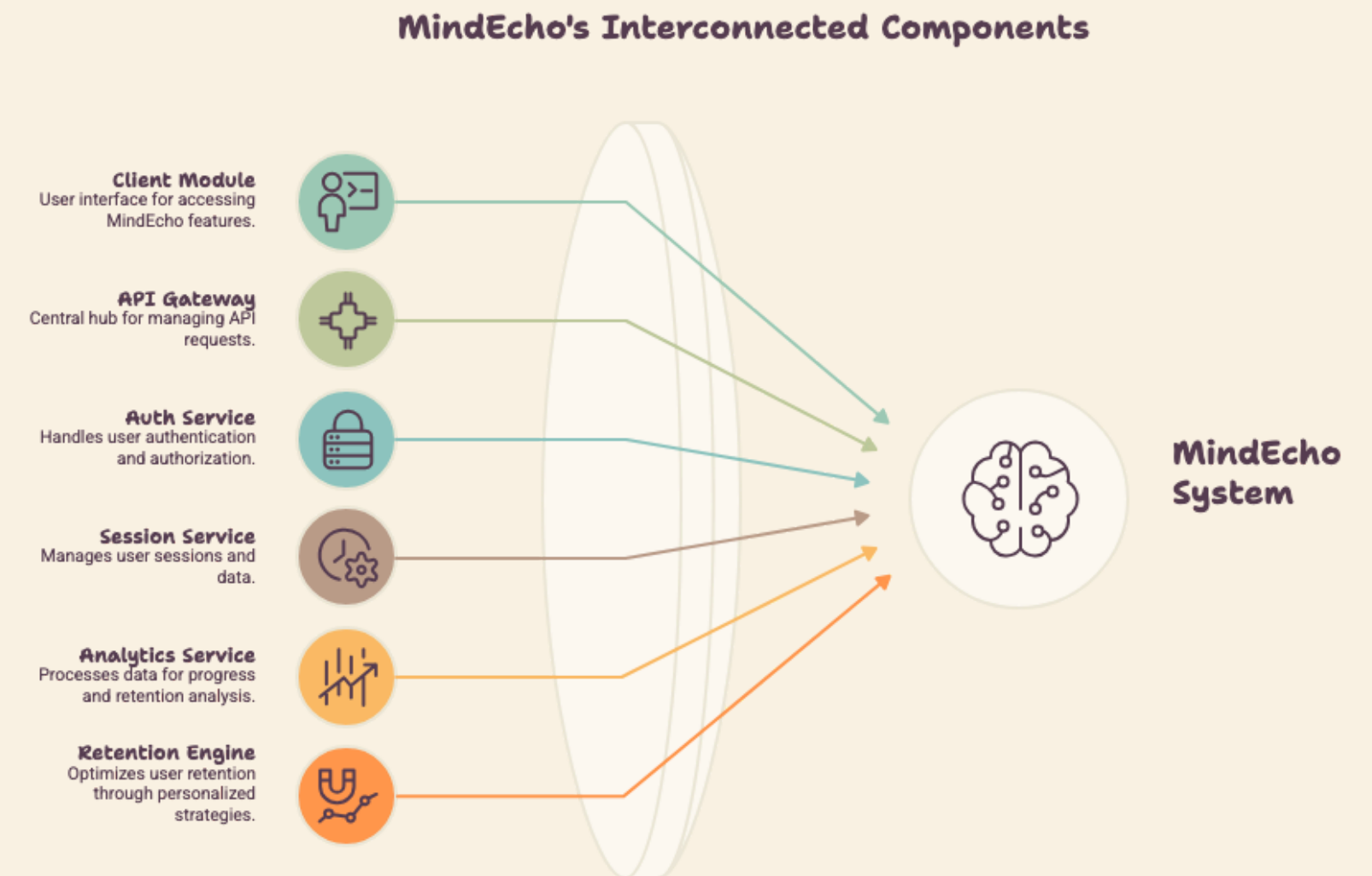
Uses a standard web technology stack with a hosted LECTOR LLM API; no GPU or local model training is required.
Uses a lightweight and explainable retention model:

$$R(t) = e^{-\frac{\lambda t}{s}}$$

Can run on laptops, smartphones, and tablets with internet access and a microphone.

CHALLENGES & RISKS

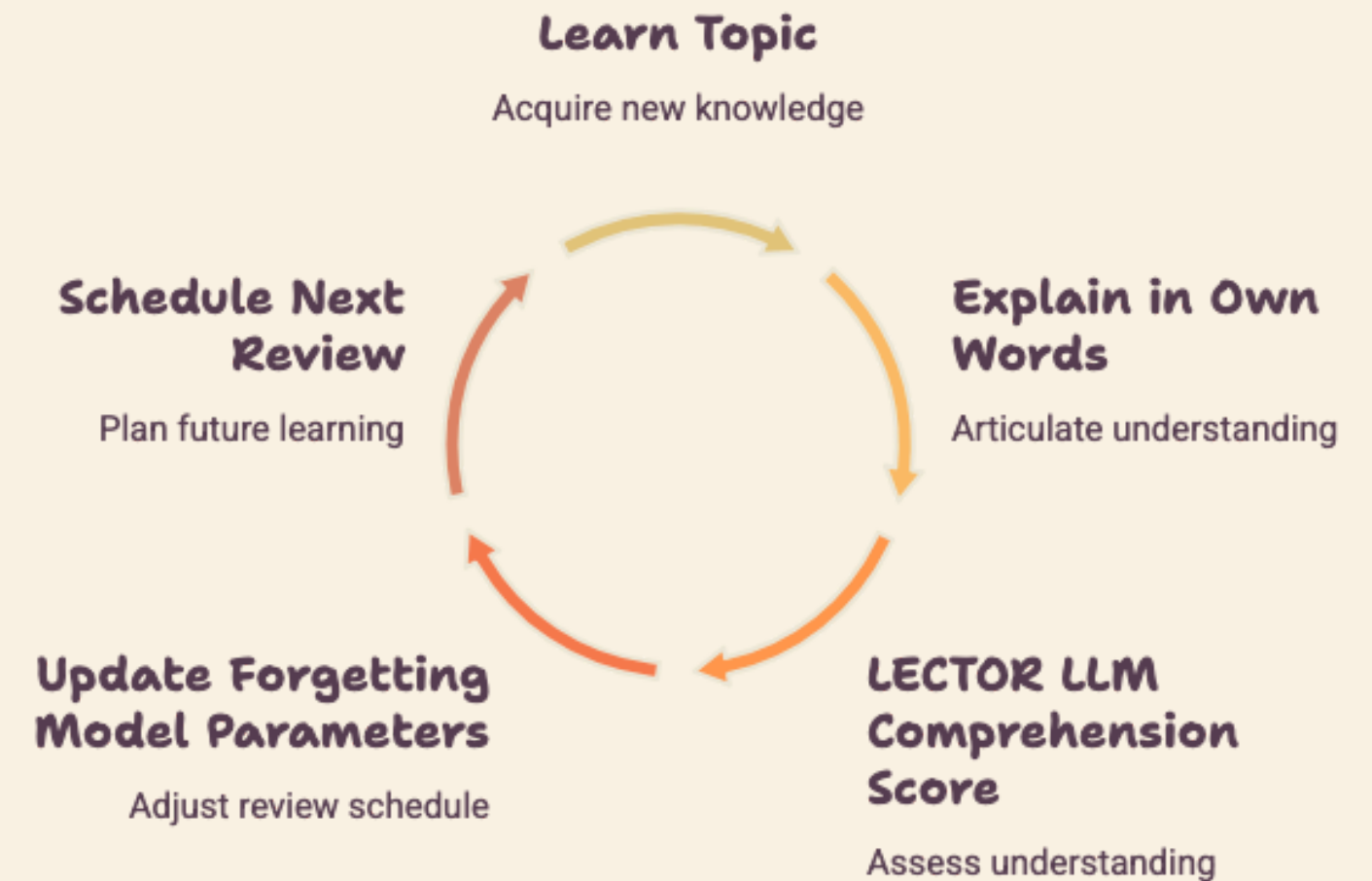
- **LECTOR LLM scoring** may vary across repeated evaluations.
- Voice capture or transcription may be affected by **background noise**.
- Incorrect **λ /s calibration** may result in review intervals that are too short or too long.
- Dependence on the **LLM API and internet** may affect system availability.



STRATEGIES TO OVERCOME

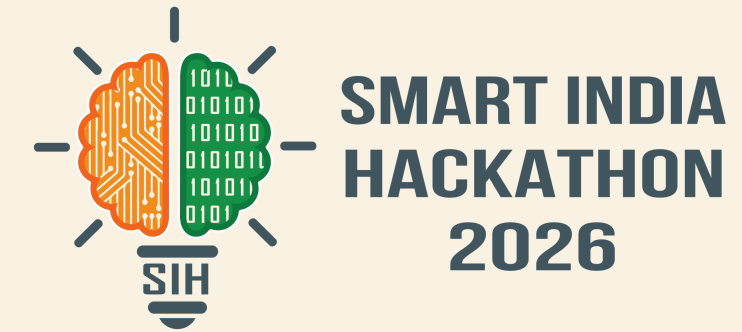
- Use a fixed 1–5 scoring rubric and perform repeated-run consistency testing for LECTOR LLM.
- Provide text input as a fallback when voice capture or transcription fails.
- Unit-test the scheduler using controlled comprehension scores and edge cases.
- Begin with a simple, explainable retention model before introducing more complex personalization.

MindEcho Adaptive Learning Cycle





IMPACT AND BENEFITS



TARGET AUDIENCE

- Students preparing for **exams, vivas and competitive tests**.
- Learners preparing for **professional certifications**.
- **Language learners** seeking better retention.
- **Corporate upskilling** and continuous-learning programs.

EXPECTED OUTCOME

- A working **Explain → LECTOR LLM Score → Update → Reschedule** learning loop.
- Adaptive review dates based on the learner's comprehension.
- A visible comparison between **adaptive revision and fixed-interval revision**.

ECONOMIC / PRACTICAL BENEFITS

- **Data-efficient personalization** reduces dependence on massive historical datasets.
- Web-based architecture avoids **specialized hardware and local model training**.
- Uses the **LECTOR LLM API** for comprehension evaluation without requiring a locally hosted model.

LEARNING BENEFITS

- Encourages **active retrieval** and explanation in the learner's own words.
- Focuses revision on **concepts that need more attention**.
- Supports **interleaved practice** across prerequisite-linked topics.
- Adapts review timing based on the learner's **comprehension and retention**.

RESEARCH AND REFERENCES

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